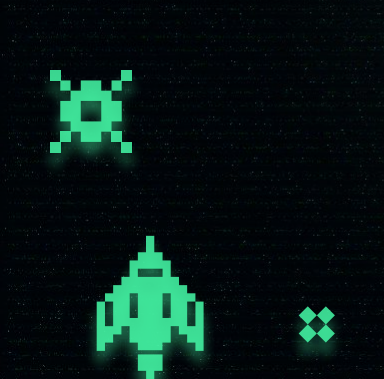




Little Lantern Stall



Alexandra Dones
April 16, 2026

Homeland Security
Final Presentation



Overview



Project concept and goal



Motivation and background



System design and
implementation



Final results and demo

Goals

01

Create 2D Cozy management game

02

Integrate cybersecurity awareness into gameplay

03

Teach security through player interaction



Motivations



Interest in cozy
simulation games



Background in
cybersecurity



Explore learning
through gameplay

Contributions



Combines cozy
gameplay with
cybersecurity
concepts



Integrates
cybersecurity risks
into everyday
gameplay situations



Fully playable and
complete game
system



Uses player decisions
to reinforce safe
behavior





¥0

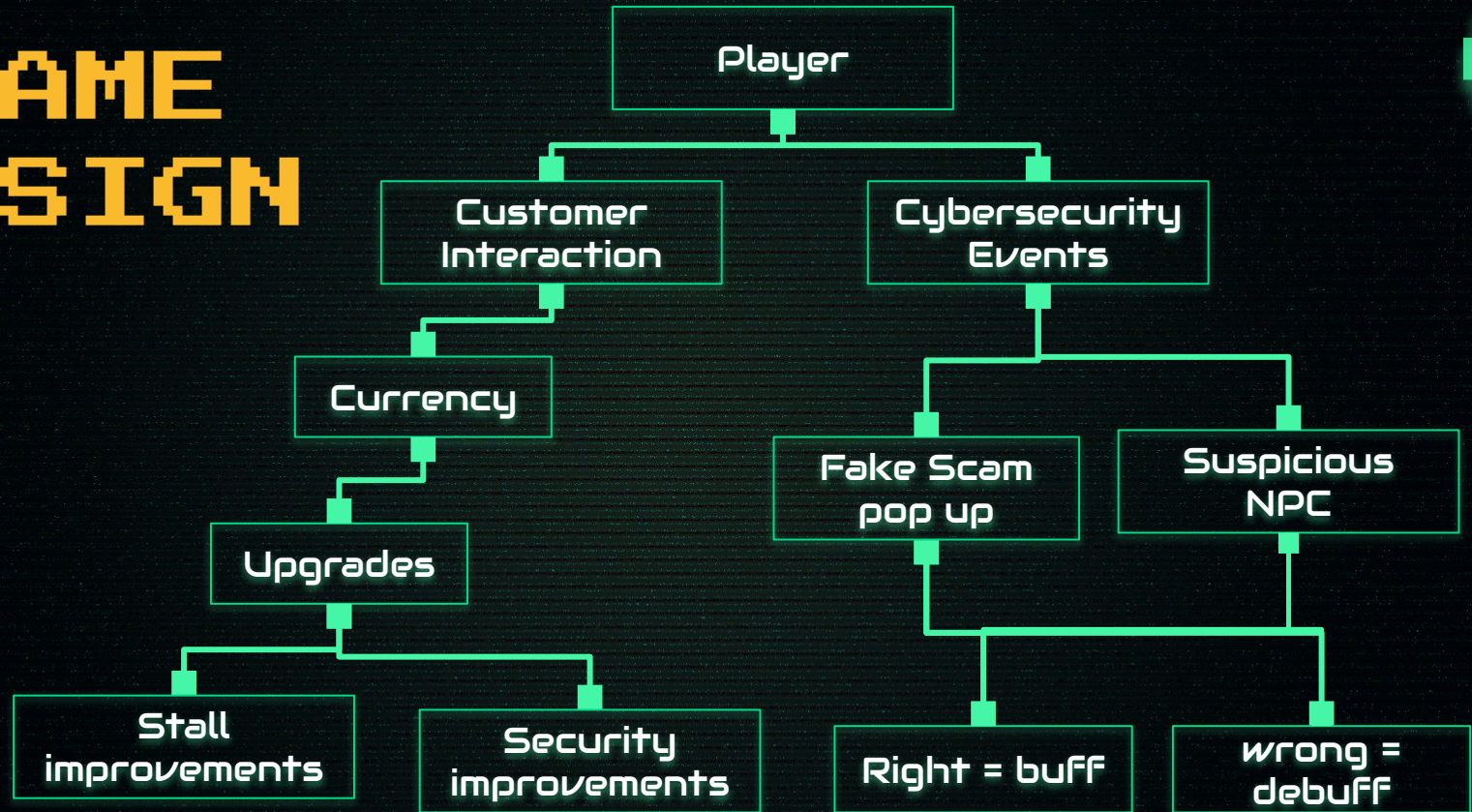
Chair 3 = ¥1500

BACKGROUND

- ★ Human behavior plays a big role in cybersecurity risks
- ★ Many attacks rely on scams and social engineering
- ★ Games can help people learn through interaction



GAME DESIGN





Unlock 4th chair -
increase customer



¥900

Chair 4 = ¥4000

Implementation

- ❑ **Customer system:**
(spawning, seating, leaving)
- ❑ **Coin system:**
(earning and collecting)
- ❑ **Upgrade system:**
(progression and unlocks)
- ❑ **Event system:**
(Sus NPC and fake popup interactions)



Cybersecurity Mechanics



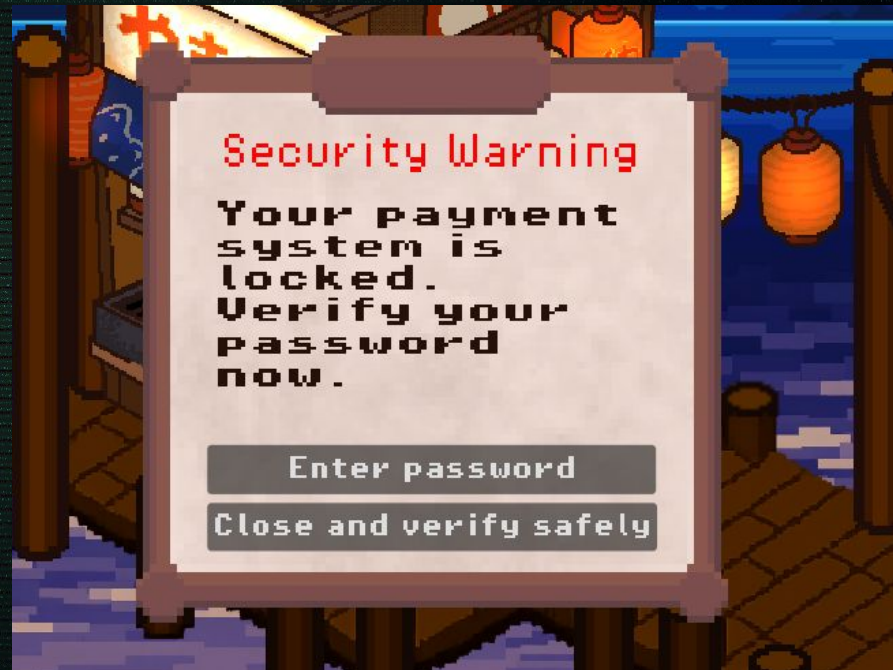
Suspicious NPC
(social engineering)



Fake popup scam
(phishing)

- ❑ Player choices create rewards or penalties
- ❑ Uses gameplay to simulate real-world scenarios
- ❑ Buffs and debuffs affect gameplay outcomes

Cybersecurity Mechanics



Cybersecurity Mechanics

```
func apply_spawn_debuff(duration: float) -> void:  
» spawn_time_multiplier = 2.5  
» effect_active = true  
» current_effect_type = "spawn_debuff"  
  
» effect_label.text = "Slow Customers ="  
» effect_panel.visible = true  
  
» customer_timer.stop()  
» start_customer_timer()  
  
» effect_timer.wait_time = duration  
» effect_timer.start()
```

Slow Customers = 49s

```
func apply_coin_buff(duration: float) -> void:  
» coin_multiplier = 2  
» effect_active = true  
» current_effect_type = "coin_buff"  
  
» effect_label.text = "2x Coins Drop ="  
» effect_panel.visible = true  
» effect_timer.wait_time = duration  
» effect_timer.start()
```

2x Coins Drop = 112s



Final Result / Cost / Time Table

- ✓ Fully playable game with complete gameplay loop
- ✓ All core systems and cybersecurity mechanics are implemented and working
- ✓ Totally Free = \$0
 - Godot and Piskel for development and art
 - itch.io for Free game assets
 - Pixabay and OpenGameArt for sound effects
- 🕒 Adjusted milestones during development
- 🕒 Some Features were removed or simplified





¥74800

Sold Out

The background is a dark blue gradient with pixel art elements. At the top center is a white skull with a wide grin. There are white pixel clouds in the top left and bottom left corners. Several white pixel stars of different sizes are scattered across the background. The word 'Conclusion' is written in a large, white, pixelated font in the upper middle section.

Conclusion

- Successfully built a complete and playable game
- Achieved project goals of integrating cybersecurity into gameplay
- Faced challenges implementing cybersecurity mechanics
- Some features were simplified to meet time constraints
- Learned game development, system design, and security concepts
- Future improvements: more events, deeper mechanics, expanded gameplay

References

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Demo

<https://youtu.be/79q6gsWjriA>
<https://kazcii.itch.io/little-lanter-stall>